Math 520, F07 **Name:** Answer

Quiz 2 **SSN:** xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (8 points) Find the general solution of the given differential equation:

$$\frac{dy}{dt} = 1 - e^t + y^2 - e^t y^2.$$

Solution: $\frac{dy}{dt} = (1-e^t)(1+y^2)$

$$\frac{1}{1+y^2}\frac{dy}{dt} = 1-e^t$$

So $f(y) = \frac{1}{1+y^2}$, $g(t) = 1-e^t$

$$F(y) = \int f(y)\,dy = \int \frac{1}{1+y^2}\,dy = \arctan y$$

$$\Rightarrow \frac{d}{dt}[F(y(t))] = g(t)$$

$$\arctan y(t) = \int 1-e^t\,dt + C$$

$$= t - e^t + C$$

$$\Rightarrow y(t) = \tan(t - e^t + C)$$

Problem 2. (12 points) Find the specific solution of the given initial-value problem:

$$2t(1+y^2) + y\frac{dy}{dt} = 0, \quad y(0) = -1.$$

Solution: $\frac{y}{1+y^2}\frac{dy}{dt} = -2t$

Here, we get $f(y) = \frac{y}{1+y^2}$, $g(t) = -2t$

$$\int_{y(0)}^{y(t)} f(r)\,dr = \int_0^t g(s)\,ds$$

$$\int_{-1}^{y(t)} \frac{r}{1+r^2}\,dr = \int_0^t -2s\,ds$$

$$\frac{1}{2}\ln(1+r^2)\Big|_{r=-1}^{r=y(t)} = -s^2\Big|_{s=0}^{s=t}$$

$$\frac{1}{2}\ln(1+y^2(t)) - \frac{1}{2}\ln 2 = -t^2$$

$$\ln\frac{1+y^2(t)}{2} = -2t^2$$

$$1+y^2(t) = 2e^{-2t^2}$$

$$y(t) = \pm\sqrt{2e^{-2t^2}-1}$$

Notice that $y(0) = -1 < 0$, so we must have.

$$y(t) = -\sqrt{2e^{-2t^2}-1}$$

The $y(t)$ is only defined on.

$$2e^{-2t^2} - 1 \geq 0$$

$$e^{-2t^2} \geq \frac{1}{2}$$

$$-2t^2 \geq -\ln 2$$

$$t^2 \leq \frac{\ln 2}{2}$$

$$\Rightarrow t \in \left[-\sqrt{\frac{\ln 2}{2}}, \sqrt{\frac{\ln 2}{2}}\right]$$